Table 19.7 Average atmospheric and aquatic releases of radionuclides from various reactor types 1990–1997

Release (kBq/kWh)	PWR	BWR	GCR	HWR	RBMK	FBR
Airborne						
Noble gases	2.3	3.0×10^{1}	1.6×10^{2}	1.3×10^{2}	1.2×10^{2}	3.4×10^{1}
^{3}H	2.7×10^{-1}	1.0×10^{-1}	4.9×10^{-1}	5.6×10^{1}	3.0	5.6
^{14}C	1.9×10^{-2}	5.5×10^{-2}	1.1×10^{-1}	3.7×10^{-1}	1.5×10^{-1}	1.4×10^{-2}
^{131}I	2.9×10^{-5}	6.3×10^{-5}	1.0×10^{-4}	2.9×10^{-5}	8.0×10^{-4}	2.9×10^{-5}
Particulates	1.7×10^{-5}	3.0×10^{-2}	2.9×10^{-5}	5.7×10^{-6}	1.3×10^{-3}	7.4×10^{-4}
Liquid						
^{3}H	2.3	1.0×10^{-1}	2.9×10^{1}	4.7×10^{1}	1.3	2.0×10^{-1}
Other	1.5×10^{-3}	3.1×10^{-3}	6.9×10^{-2}	9.9×10^{-3}	6.3×10^{-4}	4.1×10^{-3}

(Data from UNSCEAR 2000)

Legal release limits are usually set so that a critical group of the general public should at most get an insignificant additional dose, typically 0.1 mSv/y. It is also customary to put a legal limit on the collective dose (usually per GW installed generating capacity) arising from operation of a nuclear power plant.

19.17. NUCLEAR FUSION

For more than 50 years scientists have studied methods of copying the fusion energy processes occurring in the sun and other stars in which hydrogen is transformed into helium. The availability of hydrogen and deuterium in the sea is so large that it will outlast any other non-renewable energy sources for any population growth rates: 1 liter of sea water contains deuterium with an energy content equivalent to 300 liter of gasoline (reaction 12.14).

At the first Geneva conference on the peaceful uses of atomic energy in 1955 there was strong sentiment that controlled thermonuclear reactors (CTR) would be in operation within 15 years. The optimism has gradually vanished as research has uncovered numerous technical difficulties. At present, it seems quite unlikely that such reactors will be in operation before the year 2050, although extensive research is being conducted towards CTR's, and great progress is being made. The following sections briefly describe the principal concepts, and some of the problems of controlled thermonuclear (fusion) power generation.

19.17.1. The controlled thermonuclear reactor

Fusion reactions are most easily achieved with hydrogen atoms because of the low coulomb barrier and favorable wave mechanical transmission factor. The threshold

reaction energy (E_{CB}(min), eqn. (12.13)) for the $^1H + {}^1H$ reaction is 1.11 MeV, which corresponds to an average temperature of 10^{10} K; at 10^8 K the fraction of particles with an energy ≥ 1.11 MeV is about 10^{-55}, at 10^9 K about 10^{-5}, and at 10^{10} K about 0.5. The quantum mechanical tunnel effect allows the reaction to proceed at an acceptable rate at lower temperatures: For the D + T reaction the ignition temperature is 3×10^7 K and for the D + D and D + ^{3}He reactions it is 3×10^8 K. These reactions are the prime candidates for controlled fusion; see Figure 12.8. Of the number of designs proposed for CTR's the present discussion is limited to inertial confinement and magnetic confinement systems.

Several factors determine the possibility of achieving a thermonuclear fusion reaction: the particle energy, which is related to the temperature (§3.6.2), the particle density and the reaction rate. These are connected in three design criteria:

(i) the *fusion reaction rate* parameter σv, where σ is the reaction cross section (which depends on the particle energy, cf. Fig. 12.8), and v the relative speed, or temperature, of the ions, averaged over the Maxwellian velocity distribution (the temperature must be $\geq 10^8$ K),

(ii) the *Lawson limit* $n\tau$, where n is the particle density and τ the confinement time, which indicates the ability of the plasma to retain its heat and is also called the *confinement quality* — the product must be $\geq 10^{20}$ particles s m^{-3} for the DT-reaction, and $\geq 10^{21}$ s m^{-3} for the DD-reaction,

(iii) the *fusion product* $n\tau T$, where T is the average ion temperature; this product must be $\geq 5 \times 10^4$ s eV m^{-3}.

The steady state reactor is limited in power density by heat transfer and other considerations to about $n = 10^{20}-10^{21}$. Since each collision involves two particles, the fusion power density varies as the square of the particle density. At 1 Pa (i.e. 3×10^{20} particles m^{-3}) the power density would be tens of MW per m^3. This leads to a required confinement time of about 0.1−1 s.

19.17.1.1. Inertial confinement

Inertial confinement is a pulsed operation system[7]. Small pellets of solid D_2 and T_2 ($\leq$1 mm) are placed into the middle of a chamber where the pellets are irradiated by intense beams of photons (from lasers) or electrons (from accelerators). The surface of the pellet rapidly vaporizes, resulting in a jet-stream of particles away from the pellet and an impulse (temperature-pressure wave) which travels into the pellet, increasing the central temperature to $>10^8$ K. This causes a small fusion explosion, producing energetic ^{4}He and n, eqn. (12.20). Because the particle density is high, the pulse time can be very short and still meet the Lawson criterion. Temperatures of $\sim 10^9$ K have been reached with electron beams, and fusion neutrons produced. With a repetition rate of $\sim$100 pellets

[7] In the uncontrolled case of a hydrogen bomb, it is a single event.

s^{-1}, a power output of 1−10 GW would be achieved. Difficulties which have hampered the development include problems of shielding against the 14 MeV neutrons and extraction of the released kinetic energy.

19.17.1.2. Magnetic confinement

At $T = 10^7$ K, hydrogen atoms are completely dissociated into H^+ and free e^- (the plasma state). Because no construction material can withstand a plasma of this energy, it is necessary to keep it away from walls, which can be done by using strong magnetic fields, which is the same principle as used in mass spectrometers or cyclotrons. Of the various designs of "magnetic bottles", the torus (Fig. 19.17) seems to be the most promising and is the design chosen for further development in several national and international programs. The first torus machine was built in Moscow and named Tokamak; these machines have since then been called *tokamaks*.

19.17.2. Experimental fusion reactors

Several large machines based on magnetic confinement have been built. The best results so far obtained, though not simultaneously, are: central ion temperature of 35 keV (4×10^8 K), confinement time of ∼2 s, particle density of $\sim 5 \times 10^{19}$. Figure 19.18 shows the results obtained with different machines. Table 19.8 lists the results obtained with the JET machine and design parameters for ITER (International Thermonuclear Reactor Experiment). JET (Joint European Torus), the largest tokamak machine in operation, is an international project located at Abingdon, England. ITER is also an international experimental project which is expected to produce >1000 times the power of JET. ITER is being built in Cadarache, France, and it is planned to be operational

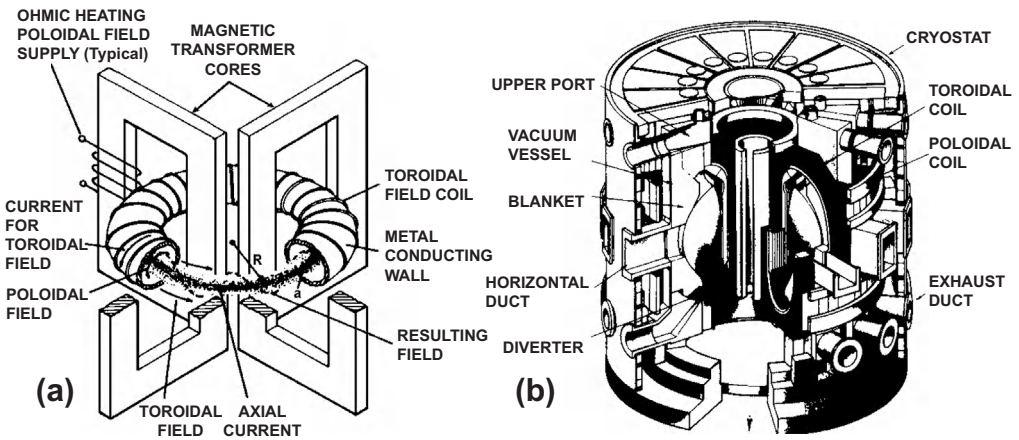


Figure 19.17 (a) The torus magnetic confinement principle. (b) Schematics of the JET (or ITER) tokamak experimental reactor.

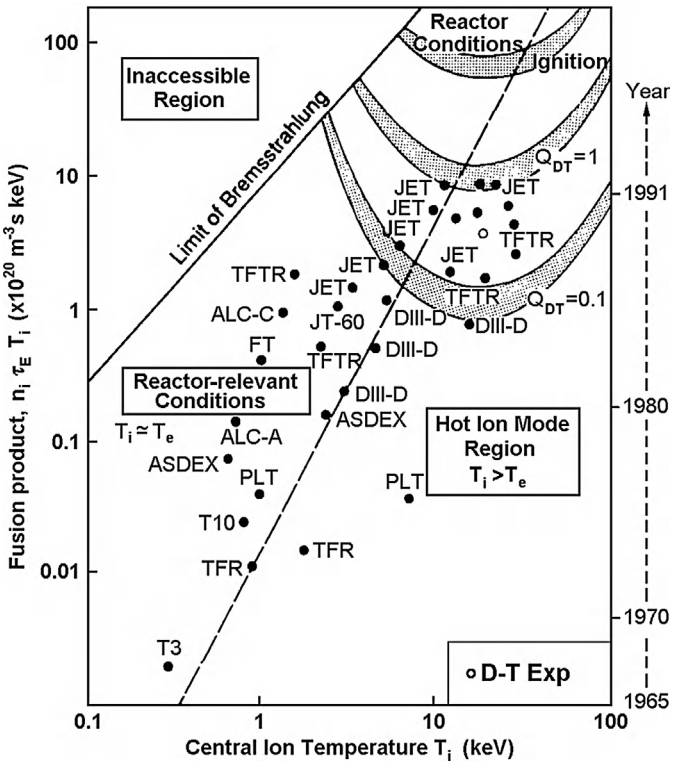


Figure 19.18 Results obtained by different fusion reactor designs.

in 2018. If promising results are obtained with ITER, it is planned to be followed by a power prototype, possibly some time after 2030 A.D.

19.17.3. Technical issues of large fusion reactors

A large fusion reactor appears to be more complicated than a fission reactor. The cost of the raw material for the two energy systems — ^{2}H- and ^{6}Li-enrichment, and T-production (~50% by irradiation in fission reactors), compared to U-production and enrichment — are probably rather similar, and is a minor factor in the net economy of both systems. Waste handling will perhaps be cheaper for the fusion system. On balance, it is unlikely that fusion energy will become much cheaper then fission energy. As long as the DT-reaction is used, fuel supply will be limited by the availability of Li and the efficiency of converting this litium to tritium. Further, considering that with fission breeders the uranium and thorium energy resources are sufficient for several centuries, there seems to be little incentive for the industry to engage in a rapid development of large fusion reactors, although a steady, well planned research is expected to continue on an international basis as fusion energy using the DD-reaction promises to be the long term answer to the world's energy needs.

Table 19.8 Parameters of Tokamak fusion reactors

	JET	ITER
Torus radius, central point (m)	3.0	6.0
Vacuum chamber: width (m)	1.2	2.05
height (m)	3.0	6.0
Total reactor height (m)	11.3	27.6
Total reactor radius (m)	17.5	31.0
Plasma current (MA)	4–7	22
Toroidal field (T)	3.5	4.9
Power in (MW)	36	
Operation:		
ion temperature T (keV)	25	10–20
" (K)	2×10^8	$\sim 10^8$
ion density, n ($\times 10^{19}$ m^{-3})	4	25
pulse length/confinement time τ (s)	1.2 (25)	400
fusion product $n\tau T$ ($\times 10^{20}$ keV s m^{-3})	8–9	50
DT reactions/s	6×10^{17}	10^{21}
T-consumption (kg/d)[†]	0.43	
DD reactions/s	1×10^{17}	
power out (thermal MW)	2[‡]	1000

[†]At continuous operation.
[‡]For about 2 s in DD reaction.

(a) *Particle injection and withdrawal.* In a steady state machine, whether it operates in pulsed mode or not, fuel (i.e. D and T) must be injected and, after consumption, reaction products (^{3}He, ^{4}He) must be withdrawn. Because of the strong magnetic field, injection of ions is an extremely difficult problem. Consequently, systems with injection of high energy neutral particles are under study. This could be accomplished through particle charge exchange, e.g.

$$D^+(\text{high energy}) + D(\text{low energy}) \rightarrow D(\text{high energy}) + D^+(\text{low energy}) \quad (19.45)$$

If all ions are removed magnetically, a beam of uncharged D-atoms (neutral beam injection, NBI) can enter the fusion volume. Another alternative being tested is high speed injection of small frozen D-T pellets into the hot plasma were they vaporize.

The withdrawal problem is even more difficult. Presently a shut down is necessary when large amounts of helium have been formed, after which the vacuum torus is purged and filled with a fresh D-T mixture. Considering the requirement of short interruptions in the energy production and the need to maintain an extremely high vacuum in the whole torus volume, a successful technical solution to this step is crucial.

(b) *Plasma confinement and heating.* The density must be increased to meet the Lawson criterion. This is achieved by increasing the magnetic field, which also leads to an

ohmic heating of the plasma, though possibly not sufficient for ignition. Additional heating is required which possibly can be achieved by high frequency (ion cyclotron resonance) heating. Injection of deuterium/tritium ions of high kinetic energy is another (cf. eqn. (19.45)). After ignition further heating comes from the 3.5 MeV α-particles of reaction, eqn. (12.20).

A magnetic field cannot, of course, increase indefinitely. When it has reached it maximum value, the machine must be shut down to allow the magnets to relax. The cycling time may be 3–10 minutes. The magnetic field goes to extremely high values, requiring superconducting magnets and extremely sturdy construction, as the high current ($\geq$30 000 A) through the magnet coils leads to a pressure of $>$1 ton cm^{-2}.

Particle injection, withdrawal, and heating lead to the emission of bremsstrahlung and synchrotron radiation of an energy much less than that corresponding to the fusion temperature. It is, therefore, lost by the plasma and absorbed in the walls of the vessel, which must be cooled. An additional difficulty is the heat insulation required between the very hot walls of the vacuum vessel ($\sim$1000°C) and the current carrying very cold super-conducting coils (at $\sim$4.5 K).

(c) *Energy extraction and fuel cycle.* In the DT cycle 80% of the energy appears as neutron kinetic energy (14 MeV). Most steady state concepts involve the capture of this energetic neutron in a surrounding blanket containing lithium as a metal or salt (e.g. Li_2BeF_4) in which tritium is produced according to

$$^{6}\mathrm{Li}\,(7.4\%) + \mathrm{n} \;\rightarrow\; {}^{3}\mathrm{H} + {}^{4}\mathrm{He} \quad Q = 4.78\ \mathrm{MeV} \tag{19.46}$$

Neutrons reacting with the other Li-isotope yield two ^{4}He:

$$^{7}\mathrm{Li}\,(92.6\%) + \mathrm{n} \;\rightarrow\; 2\,{}^{4}\mathrm{He} \quad Q = 18.13\ \mathrm{MeV} \tag{19.47}$$

These reactions develop additional kinetic energy, which is converted into blanket heat. Most of the energy of the fusion process appears therefore in the hot blanket. If the blanket is in the form of molten lithium metal or salt, it can be pumped through a heat exchanger, which, in a secondary circuit, produces steam for a turbine. Although the melting point of Li metal is only 186°C the Li temperature in the system will probably be around 1000°C. The tritium produced must be recovered for recycling as fuel.

The DT reactor needs several kg tritium as starting material (1kg ^{3}H $\sim$ 3.6 $\times$ 10^{17}Bq). A likely technique involves the irradiation of a ^{6}Li-Al alloy in a high flux thermal fission reactor which produces both tritium and ^{4}He, see eqns. (19.48) and (19.49)); These can be separated on the basis of their different vapor pressures, different permeability through palladium, or through their different chemical reactivities.

The first fusion reactors probably will use the DT-reaction, as the DD-reaction requires higher temperatures, Figure 12.8. The DT-reaction yields on the average 0.5 neutrons. Provided the blanket consists of only ^{6}Li, this produces 0.5 new T-atoms (per consumed T-atom); if the blanket also contains ^{7}Li, the yield of new T will be less. Therefore, the DT-fusion reactor must be fed continually with new tritium, produced in fission reactors. This demand will not be eliminated until the DD-fusion reactor comes into operation.

(d) *Construction material.* A major area of research is the proper choice of construction material, which should be strong, heat and radiation resistant, and have low neutron capture cross sections. However, materials testing will not be intensely tackled until ITER comes into operation. A preliminary materials choice could be the following, starting from the center of the plasma:

(1) A vacuum chamber of 10 cm stainless steel, on the inside lined with reinforced carbon shields to protect the steel. The chamber walls are water cooled. As the shielding erodes it has to be occasionally replaced by remote control.

(2) A water cooled blanket (and shield), ca. 1.5 m thick, made of vanadium alloy (e.g. V-15Cr-5Ti) and containing solid or liquid lithium. As much heat is produced, the lithium is used both for cooling and breeding and is pumped to heat exchangers for steam production followed by chemical treatment to recover the tritium. Cooling can also be achieved by He-gas, in which case the Li-system must be treated separately.

(3) A water cooled shield and thermal insulator, containing γ- and n-absorbing material.

(4) A liquid helium cooled superconducting magnet coil.

(e) *Health, environmental and economic aspects.* An operating reactor would contain several kilograms of tritium (1 kg of T_2 is about 3.6×10^{17} Bq of radioactivity) which presents a hazard corresponding to the noble gas fission products. However, tritium is more difficult to contain because of its ability to pass through many metals, especially at high temperatures. Moreover, tritium can exchange with hydrogen atoms in water, and thus become an inhalation and ingestion hazard. Tritium is already a problem for fission reactors; for fusion reactors the problem is at least a factor of 1000 greater. Hence, the fusion reactor must be sealed extremely well against tritium leakage.

The structural material will be exposed to high radiation fields, causing radiation damage and induced radioactivity. The preferred material at present seems to be reinforced carbon, special steel and vanadium. Thus, considerable amounts of ^{49}V ($t_{½}$, 330 d), and possibly also some very long-lived ^{53}Mn, are formed. This induced activity will be a maintenance hazard, requiring remote control systems. However, compared to a fission reactor of similar size, the fusion reactor will contain less total radioactivity, and (of special importance in waste disposal) be free of long-lived α-activities.

A safety aspect, which has not been thoroughly studied, is the rapidly changing strong magnetic field (1 T s^{-1}), which will put great stress on the coils and structural material. The effects of such high fields on the operators are unknown.

19.18. NUCLEAR EXPLOSIVES

Assume that 50 kg of metallic ^{235}U is rapidly brought together to provide a critical configuration. Using a generation time of 3×10^{-9} s for neutrons, it can be calculated with eqns. (19.33) and (19.34) that it would take 0.2×10^{-6} s to increase the number of neutrons from one to that required for fissioning all ^{235}U atoms. However, long before this time, the energy absorbed in the material would have blown it apart, so 100% efficiency is impossible in a nuclear explosion. If 2% of the 50 kg of uranium had been fissioned, the amount of energy released corresponds to about 20 000 t of TNT (1 t of TNT corresponds to 4.19×10^9 J). The energy production is said to be 20 kt. Only ^{233}U, ^{235}U, and ^{239}Pu (of the more easily available long-lived nuclides) have sufficiently high cross sections for fission by fast neutrons to have a reasonable critical mass for use as nuclear explosives. However, it is a common misconception that when a critical mass is reached, a nuclear explosion will follow. In the cases when a critical mass has been reached by mistake, no energetic nuclear explosion has ever occurred.

A rough estimate of the critical radius of a homogeneous un-reflected reactor may be obtained simply by estimating the neutron mean free path according to (11.6). Assuming ^{235}U metal with a density of 19 g cm^{-3} and a fast fission cross section of 2×10^{-24} cm^2, one obtains $\Sigma_f^{-1} = 10$ cm. A sphere with this radius weighs 80 kg. For an un-reflected metal sphere containing 93.5% ^{235}U the correct value is 52 kg. ^{239}Pu has the smallest un-reflected critical size; for ^{239}Pu (δ-phase, density 15.8 g cm^{-3}) it is 15.7 kg (~6 kg reflected), and for ^{233}U 16.2 kg (~6 kg reflected).

Assume two half-spheres of metallic ^{235}U, each of which is less than critical size but which together exceed criticality. As the distance between the two spheres is diminished the k_{eff} increases for the total system. At a certain distance k_{eff} can be maintained close to 1 but the chain reaction still be prevented. If the distance between the two half-spheres is diminished rapidly to less than this critical distance in the absense of any neutron, a single neutron can then trigger a rapidly multiplying chain reaction. Since stray neutrons are always available, e.g. from spontaneous fission, a chain reaction would be started when the distance between the spheres is less than the critical distance. Hence it is important to achieve the supercritical configuration fast compared to the average time interval between the stray neutrons.

In order to achieve a substantial energy release in the short time available before the supercritical configuration is blown apart, an external or internal neutron source is triggered. By this action the fission chain reaction starts with not a few, but with very many high energy neutrons. This greatly increases the number of neutrons generated in